

RAWSON LAKE, ON, Canada

WMO: 718550

Lat: 49.660N

Lon: 93.728W

Elev: 1427

StdP: 13.95

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.3	-18.8	-29.7	1.1	-22.9	-24.9	1.5	-18.6	13.2	11.6	12.1	12.2	3.6	230	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	83.6	67.2	80.6	65.5	77.8	63.8	70.8	78.4	68.9	76.2	66.9	74.1	7.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	109.5	74.8	66.3	102.0	73.3	64.3	95.1	70.7	35.7	78.5	34.0	75.9	32.3	74.3	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.5	11.9	10.7	-28.0	88.7	3.9	2.3	-30.8	90.4	-33.0	91.7	-35.2	93.1	-38.0	94.8		
			-28.5	73.8	3.7	1.6	-31.2	74.9	-33.4	75.9	-35.5	76.8	-38.2	77.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.9	6.0	6.1	24.3	38.0	51.3	61.5	67.3	64.8	54.8	41.4	26.4	10.4	(6)
	DBStd	24.08	14.17	12.21	12.99	11.15	9.03	5.95	5.49	5.59	8.36	8.81	11.06	12.82	(7)
	HDD50	6143	1364	1230	800	378	95	2	0	0	42	297	708	1227	(8)
	HDD65	10133	1829	1650	1262	811	431	135	39	73	321	732	1158	1692	(9)
	CDD50	1719	0	0	4	16	136	347	537	460	187	31	1	0	(10)
	CDD65	230	0	0	0	0	7	30	111	67	15	0	0	0	(11)
	CDH74	1621	0	0	0	0	70	231	778	430	110	2	0	0	(12)
	CDH80	312	0	0	0	0	6	30	182	80	14	0	0	0	(13)
	WSAvg	5.4	5.2	5.1	5.5	5.9	5.9	5.2	5.1	5.0	5.4	5.6	5.6	4.9	(14)
	PrecAvg	26.9	1.2	0.9	1.6	1.6	2.6	4.1	3.4	3.4	3.4	2.2	1.9	1.1	(15)
(16) Precipitation	PrecMax	34.3	2.7	1.5	3.1	4.4	5.7	6.8	7.1	6.6	6.5	5.8	6.2	2.2	(16)
	PrecMin	20.0	0.2	0.2	0.2	0.0	0.8	1.8	0.8	1.1	0.6	0.7	0.6	0.8	(17)
	PrecStd	4.6	0.6	0.3	0.7	1.3	1.4	1.7	1.8	1.4	1.7	1.4	1.2	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	36.0	63.6	69.1	80.6	83.9	88.7	86.2	82.1	71.0	55.6	38.4	(19)
		MCWB	34.2	31.2	54.2	51.0	60.9	66.7	69.6	69.1	65.0	58.1	47.4	34.7	(20)
	2%	DB	32.8	31.9	52.5	63.5	75.8	79.6	84.7	82.1	77.3	65.3	48.0	32.2	(21)
		MCWB	31.8	29.3	44.8	47.9	57.9	63.0	67.8	67.5	63.8	55.3	42.3	31.0	(22)
	5%	DB	29.6	28.2	46.2	59.0	71.6	76.4	81.5	78.5	72.4	59.3	44.0	30.4	(23)
		MCWB	28.1	26.3	39.6	45.8	55.7	61.5	66.5	65.7	61.5	51.2	40.5	29.0	(24)
	10%	DB	26.1	24.1	40.8	54.7	66.7	73.4	78.3	75.7	68.2	54.9	40.6	27.9	(25)
		MCWB	24.9	22.3	35.4	43.7	54.0	61.0	64.9	64.4	59.2	48.9	37.9	26.7	(26)
	0.4%	WB	34.7	33.2	56.3	53.1	64.8	72.2	74.0	72.2	69.4	59.1	48.2	34.4	(27)
		MCDB	36.4	35.3	59.3	63.8	74.3	78.7	83.3	80.7	76.6	68.5	53.9	37.7	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	29.7	44.8	49.6	61.7	68.0	71.5	70.0	66.6	55.8	43.9	31.6	(29)
		MCDB	32.9	31.3	52.2	58.7	70.0	74.3	78.8	77.8	72.8	64.3	46.8	32.4	(30)
	5%	WB	28.4	26.0	39.8	47.1	58.5	64.4	69.3	68.0	63.3	52.5	40.8	29.2	(31)
		MCDB	29.7	28.3	45.1	57.0	66.5	71.8	76.8	75.7	70.5	57.8	43.6	30.2	(32)
	10%	WB	24.9	22.4	35.8	44.4	55.7	62.2	67.4	66.1	60.3	49.2	37.6	26.7	(33)
		MCDB	26.2	24.3	40.0	53.9	64.4	69.9	74.6	73.4	66.8	54.2	40.7	27.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	18.1	18.8	19.2	18.8	17.3	17.8	17.2	15.5	12.1	11.1	12.3	(35)
		MCDBR	14.9	17.2	20.2	24.4	25.0	22.3	21.6	21.3	21.3	19.7	14.8	11.4	(36)
		MCWBR	13.5	14.0	12.7	14.1	12.5	11.0	9.3	10.0	11.0	11.9	10.4	9.9	(37)
	5% WB	MCDBR	13.7	15.7	18.4	21.7	20.0	18.0	18.6	18.3	18.6	17.3	13.9	10.7	(38)
		MCWBR	12.7	13.1	12.2	13.5	12.2	12.0	10.4	10.5	10.9	11.3	10.1	9.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.254	0.261	0.286	0.330	0.345	0.365	0.380	0.363	0.343	0.298	0.275	0.248	(40)
	taud		2.346	2.332	2.394	2.430	2.457	2.427	2.394	2.476	2.508	2.608	2.523	2.401	(41)
	Ebn at Noon		258	285	295	291	290	284	277	277	272	271	251	246	(42)
	Edh at Noon		21	27	30	33	34	35	36	31	28	21	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		381	761	1160	1548	1693	1761	1834	1535	1113	641	341	272	(44)
	RadStd		44	68	95	121	113	144	124	125	100	87	40	34	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page